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Managing the variability of a logistics robotic system ☆

Kentaro Yoshimura [a,*], Yuta Yamauchi [b], Hideo Takahashi [c]

[a] Research & Development Group, Hitachi, Ltd., Ibaraki, Japan
[b] Control System Platform Division, Hitachi, Ltd., Ibaraki, Japan
[c] Digital Engineering Business Division, Hitachi, Ltd., Kanagawa, Japan

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ABSTRACT

This paper reports an industrial experience of managing the variability of a logistics robotic system. Logistics centers need to dispatch products rapidly and accurately from huge inventories in accordance with the daily flow of orders, and autonomous logistics robotic systems provide a solution that can handle a wide variety of products without relying on manual tasks. However, the development of such systems can be challenging due to the many different operating and product characteristics that must be considered. We have therefore developed a variability management approach that models the variability of robotic systems and the items to be handled in such a way that the software components can be reused and expanded across different products and environments. This paper presents the motivation, challenges, and benefits of applying the variability management approach to a logistics robotic system.

1. Introduction

The sharp uptick in e-commerce and home delivery demands in the wake of COVID-19 has led to a significantly increased distribution volume in the logistics industry, causing problems such as rising logistics costs and delays (Liu et al., 2021). In particular, logistics center operations such as inbound and outbound operations are facing a high demand for automation due to requirements such as handling heavy objects, night-time operations, and accurate loading. With the growth of autonomous control technology, especially robot technology, the development of automated warehouses, automated guided vehicles (AGVs), and picking robots is progressing, and research into the practical usage of automated transport robots, automated warehouses, and picking robots is picking up speed (Pikner et al., 2021; He et al., 2022; Tong et al., 2020).

Some companies have been actively advancing the research, development, and commercialization of logistics robotic systems (Endo et al., 2021). Fig. 1 shows a piece-picking robotic system we developed to automate the picking operations in a warehouse. It comes with sensing, planning, and handling functions and is responsible for transferring items from storage containers to shipping containers. A significant technical challenge here is that, in contrast to conventional logistics robots that deal with standardized shapes such as pallets, cardboard boxes, shelves, and containers, piece-picking robots need to handle a diverse range of individual products or "pieces" that vary greatly. These robots must also be able to recognize, transport, and stack new products or packages that are constantly being added and updated (Zeng et al., 2018). Another challenge is the diversity of the robotic system configurations: the robotic system layout, the surrounding material handling equipment, and the robot devices (cameras, arms, hands, etc.) must be customized in accordance with the warehouse environment of each customer. Therefore, a high degree of configurability is required for the robot control system (García et al., 2023).

To meet these customization requirements, we are moving towards the adoption of variability management, which is expected to help companies deliver variant-rich quality products quickly and at scale.

In this paper, we report the experience of applying our variability management approach to a logistics robotic system. We aim to enhance the empirical comprehension of the distinct features and obstacles in the logistics robotics field concerning variability. We expect that our results will assist practitioners and researchers in increasing the awareness of variability management's efficacy and direct subsequent studies in this domain. Our experience demonstrates how variability management can effectively address the challenges of diverse product handling and dynamic environments in logistics robotic applications. We also clarify future challenges with software product lines for autonomous systems in a variant-rich operation environment.

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* Corresponding author.
E-mail addresses: kentaro.yoshimura.jr@hitachi.com (K. Yoshimura), yuta.yamauchi.kj@hitachi.com (Y. Yamauchi), hideo.takahashi.qa@hitachi.com (H. Takahashi).

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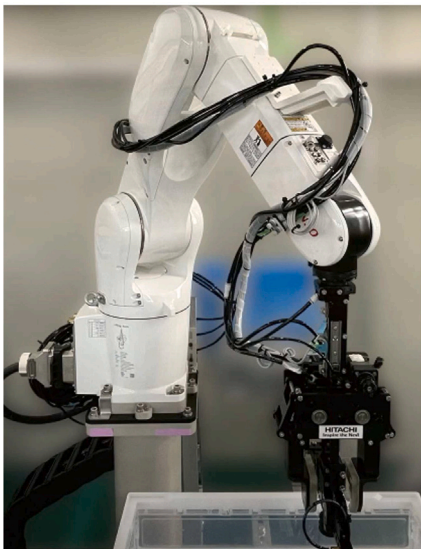

Fig. 1. Piece-picking robot for logistics.

Section 2 of this paper presents the background of the piece-picking robotic system, and Section 4 explains our motivation for adopting the variability management approach. Related work is presented in Section 3, and in Section 5, we discuss our approach to manage variability in the robotic system. Section 7 presents the insights gained from our experience and touches on future work. We conclude in Section 8 with a summary.

2. Background

The rapid growth of the e-commerce industry has resulted in an increasing number of items handled by logistics centers and created new demands for living practices. If no changes to the current situation are made, the industry will face tremendous difficulties in balancing service levels and demand. One of the main challenges is how to automate the task of picking, which involves handling individual products with diverse characteristics and work patterns. Manual labor is still preferred due to its adaptability and productivity, but future labor shortages and higher volumes of product will require alternative solutions that utilize industrial robots.

To address this issue, we have developed an innovative autonomous piece-picking robotic system that can flexibly adapt to new products and work patterns through the implementation of software-controlled robot movements. A key feature of this system is that the software can be upgraded with the latest algorithms, thus enabling it to handle new types of products or tasks with minimal development effort while using the existing software implementation. This feature enhances the efficiency and flexibility of the system. Our autonomous piece-picking robotic system thus offers a promising solution for logistics centers that need to cope with increasing demands and changing environments.

Industrial robots require a preliminary learning process in which they receive detailed instructions for joint movements related to angles, directions, and distances (Argall et al., 2009). This process necessitates specialist expertise and is time-consuming. However, once completed, robots exhibit excellent repeatability and can execute operations with high accuracy and speed. Consequently, they are widely utilized in industries such as automotive manufacturing where large quantities of products with consistent quality are required. The extended operation of production lines allows for the recoupment of initial investments in setup work and capital equipment.

In contrast to manufacturing environments, logistics warehouses present constantly varying conditions due to fluctuating order volumes and inconsistent positioning of goods within storage cases. The nearly infinite possible patterns of picking work render teaching an impractical solution for conventional robots. As a result, little progress has been made in replacing human labor in these settings.

The autonomous piece-picking robotic system, which utilizes control software to operate without teaching, is our approach to addressing the limitations of conventional robots in logistics warehouses. The software allows for the selection and integration of hardware components such as robotic arms, cameras, and end-effectors based on the specific operating conditions of a customer's facility.

The primary objective of this system is to have robots perform picking work in a manner similar to human workers. Our control software achieves this through three key functions: object recognition, handling, and transfer. Each function must manage various aspects of logistics operations.

The object recognition function involves the use of 3D camera images to determine product type, position, and distance. The system can recognize products even if they are out of position within the camera's field of view. The camera's placement may differ based on the installation facility's layout, and certain robots require a hand camera to discern package designs. Additionally, there is a variety of camera devices available.

The handling function encompasses a diverse range of product packaging and necessitates the selection and utilization of suitable end-effectors. Product packaging varies across different logistics industries, including food (e.g., cookies, beverages), medicine, cosmetics, and others. The choice of a robot's end-effector should be based on the product's properties, such as package shape, material, and weight.

The transfer function produces a series of motions that enable the robotic arm to reach and grasp an item at its gripping point, then transport it to a specified location. The calculation of the robotic arm's trajectory waypoints should consider factors such as the installation layout, the size of the robotic arm, and the packaging and placement location (e.g., tray or box).

To achieve the above concept, we implemented a variability management approach in the robotic application. We conducted three studies summarized in Table 1 to understand how the proposed approach is beneficial for the robotic application.

This enables us to

- model the variability of the robot configuration,
- model the variability of the product to be handled,
- design software architecture for the control systems,
- design dynamic behavior of the robot operations, and
- introduce the concept of core asset development and application development.

In this paper, we focus on our approach and experience with the variability management.

3. Related work

Variability management is a core process of software product line (SPL) engineering, which is an approach to improve the reusability and quality of software for variant-rich systems (Clements and Northrop, 2002; Pohl et al., 2005; Apel et al., 2013). The SPL approach is widely utilized in industrial applications such as automotive systems and railway control systems (Rabiser et al., 2018; Azanza et al., 2021; Abbas et al., 2020; Eggert et al., 2022). Inspired by the success cases in the SPL community, we endeavor to apply the SPL approach to the robotics domain.

The development of SPL technology has been a topic of interest in the robotics domain since the 2000s. Kim et al. proposed a re-engineering approach for the software architecture of home service robots as a use case (Kim et al., 2005), and more recently, with the technological progress of sensing and task planning methods, service robots are gradually being installed in our society, with examples

Table 1
Summary of studies included in this paper.

Study 1: Identifying Industrial Challenges	Study 2: Introducing Variability Management	Study 3: Conducting Interview with Experts
Approach: Market analysis and discussion with sales division	Approach: Introducing SPL approach into the existing application with SPL experts	Approach: Conducted interviews with the architect and an programmer of the robotic application
Key Finding: Domain experts reported spending time on robot learning for installation and maintenance to adopt items to be handled.	Key Finding: Variation points of robot hardware and software are dynamically selected based on its mission and environment.	Key Finding: The experts accepted efficiency of robot learning (configuration) process based on the variability management. Integration with SPL approach and robotic software development process is a future work.

including autonomous cleaning robots and automatic guided vehicles in warehouses (Ingrand and Ghallab, 2017). Due to the variation of tasks and operation environments in the robotics domain, the variability management of service robots has become a key issue in terms of practical applications (García et al., 2020). Garcia et al. studied the variability of service robot configuration based on recent applications (García et al., 2023) and found that the configuration of not only robotic devices but also service robot tasks has become more complex. Gherardi et al. proposed an SPL approach to configure robotic applications in cloud environments (Gherardi et al., 2014). To the best of our knowledge, this paper is the first to analyze and manage variability in piece-picking robots for logistics applications.

Dynamic Software Product Lines (DSPLs) have emerged as a promising approach to configure the variability of a product line during runtime (Capilla et al., 2014). In recent years, the introduction of autonomous systems, such as robotics, for adapting to changing environments and missions has sparked interest in the research field of dynamic SPL (Brugali, 2023, Ingrand and Ghallab, 2017, Srivastava and Kumar, 2018, Päßler et al., 2023, Damasceno et al., 2021). Our approach builds upon these recent research findings. Leveraging the concepts and methodologies of dynamic SPL for adaptive systems, we aim to investigate and model the variability of self-adaptive logistics robots. Furthermore, we intend to design both dynamic variability and static (less-dynamic) variability for logistic missions.

4. Challenges

To understand the challenges associated with the installation of robotic applications in logistics warehouses, we conducted a comprehensive market analysis and held in-depth discussions with salespeople as domain experts. These efforts provided valuable insights into the specific obstacles that need to be addressed to facilitate the effective deployment of robotic systems in logistics environments. The following key challenges were identified:

4.1. Diverse range of items

Unlike industrial robots used for factory assembly, logistics picking robots need to handle a wide variety of items. Imagine the number of different products in a supermarket; logistics robots must be capable of managing this extensive range of items. The burden of robot learning to handle such a large number of items is significant, making the introduction of robots challenging. This is because items may have different shapes, sizes, weights, and packaging materials, requiring the robot to be highly adaptable and versatile.

For instance, while logistics robots that only need to handle standardized items like "cardboard boxes" or "logistics pallets" have seen relatively more progress in their deployment, the scenario is quite different for piece-picking robots. These robots must be able to pick individual items for a mixed assortment, which is far more complex. To advance the application of robots in piece-picking tasks, it is crucial to reduce the labor involved in robot learning. This involves developing more sophisticated algorithms and a robot learning process that can generalize across a wide variety of items, thereby reducing the need for extensive manual programming.

Our approach addresses this challenge by adopting feature modeling (variability analysis) to systematically capture the diverse range of items and analyze their relationship with the features of the robotic system. By creating a structured model of item variability, we can better understand how different item characteristics interact with robot capabilities, allowing for more efficient configuration and adaptation of the robotic system. This method builds on the foundational work of Kang et al. (2002) and Apel et al. (2013) in feature-oriented domain analysis, extending it to the logistics robotics domain to fill a gap in current research.

4.2. Continuous addition of new products

Furthermore, the items targeted for piece-picking are constantly being updated with new products. In a dynamic market environment, new products are introduced frequently, each with unique specifications such as dimensions, weight, and packaging material. Adjusting the robot control software every time a new product is added leads to increased operational costs for the robots. This is because each new product may require reprogramming or relearning of the robot to handle it effectively.

Therefore, it is essential to enable the robots to autonomously configure their actions based on the specifications of the new products. This capability would significantly reduce the need for manual adjustments and lower the overall cost of operating the robots. By incorporating advanced sensing and adaptive control algorithms, robots can be made more intelligent and capable of self-adjusting to new items without human intervention. This would not only improve efficiency but also make the system more scalable and adaptable to future changes.

Our research contributes to this area by developing a dynamic configuration framework that leverages real-time data to adjust robot actions. This approach is aligned with the findings of García et al. (2023), who emphasized the importance of dynamic adaptability in robotic systems. Our framework extends their work by providing a practical implementation that has been validated in industrial settings.

4.3. Avoiding the clone-and-own approach

Another significant challenge is the deployment of robots across different warehouses. If a clone-and-own approach is taken, where each robotic system is cloned and slightly modified for each new warehouse, it would result in the need to manage a large number of similar but different software versions. This state is highly undesirable as it complicates maintenance and updates. The clone-and-own approach may work when the number of robot installations is small (a few cases), but as the business expands, it can lead to a substantial increase in maintenance costs.

To avoid this, a more systematic approach to software variability management is required. This involves creating a core software platform that can easily be configured and extended to meet the specific

needs of different warehouses without duplicating the entire system. By using techniques such as software product lines (SPL) and modular design, it is possible to manage variability more effectively and reduce the overhead associated with maintaining multiple versions of the software.

Additionally, we implement a dynamic reconfiguration framework to further mitigate the clone-and-own problem. This approach allows the robotic system to adapt to different warehouse environments in real-time, thus maintaining a single software version that can be flexibly configured for various operational contexts. The concept of dynamic reconfiguration has been explored in earlier works, such as Capilla et al. (2014) and Päßler et al. (2023), which emphasize the importance of runtime adaptability in complex systems. By integrating these principles, our framework not only addresses the clone-and-own issue but also enhances the flexibility and scalability of logistics robotic systems.

4.4. Summary

In summary, the challenges identified through our market analysis and discussions with salespeople highlight the need for advanced solutions to manage the variability and complexity of logistics robotic systems. Addressing these challenges involves reducing the engineer involved in robot learning, enabling autonomous configuration of robots for new products, and avoiding the pitfalls of the clone-and-own approach. By tackling these issues, we can make the deployment of robotic systems in logistics warehouses more feasible, cost-effective, and scalable, ultimately enhancing the efficiency and flexibility of logistics operations.

Our research contributes to the literature by providing empirical evidence of the effectiveness of our variability management approach in addressing these challenges. By bridging the gap between theoretical frameworks and practical applications, we offer a comprehensive solution that enhances the deployment and operation of logistics robotic systems.

5. Solution

Our objective in introducing variability management is to effectively manage the variability of robots and the products to be handled for logistics applications. Drawing on past experience with introducing SPL (Yoshimura et al., 2011), we define an adoption approach based on industrial best practices. This section provides an overview of our approach, which comprises scoping, variability analysis, defining variability mechanisms, developing core and variable components, and deriving different applications. In the following subsections, we discuss each of these items in detail.

5.1. Scoping

Our initial step towards variability management is defining the scope of the product line. Given the broad range of applications in logistics automation and service robotics, it is necessary to establish the boundaries of our variations in order to improve software reusability and quality. Through consultations with business experts, we have determined that our scope will encompass piece-picking robotic applications for logistics in the wholesale and retail domains.

5.2. Variability analysis

In the first step of variability analysis, we identify the variability of logistics robots and their types. Table 2 presents the variability of piece-picking robots for logistics, including categories, variation points, decision timing of variability, and whether the variability is static or dynamic.

The variability of logistics robots can be classified into three main categories: Environment, Hardware, and Mission.

Table 2
Variability of logistics robots.

Category	Variation point	Decision timing	Static/Dynamic
Environment	Warehouse layout	Equipment installation	Static
	Lighting	Time, Other machine	Dynamic
	Other machinery	Equipment installation	Static
	Human inclusion	Mission definition	Dynamic
Hardware	Robotic	Equipment installation	Static
	End-effector	Selected for item to handle	Dynamic
	Sensor (camera etc.)	Equipment installation	Static
Mission	Item	New item release	Dynamic
	Item layout	Order list receive	Dynamic

- Environment refers to the variability outside of logistics robots.
  - Warehouse layout varies for each warehouse and the robot's installation point within the warehouse. The layout includes the arrangement of conveyors, shelves, etc. This variation point is decided during equipment installation, making it static.
  - Lighting conditions change based on the time of day and the effects of other machinery. This variation point is dynamic, as it changes during runtime.
  - Other machinery includes specific types of conveyors and other logistics robots (e.g., AGVs). This variation point is decided during the installation of other machines, making it static.
  - Human inclusion refers to the variability of logistics robots when collaborating with human workers for tasks such as picking and packaging items for shipment. As human behavior varies for each mission and order list, this variation point is dynamic.
- Hardware refers to the variability of the logistics robots themselves.
  - The robotic arm is selected based on mission requirements (e.g., item size, weight) and warehouse layout. The selection of the robotic arm is done during equipment installation and rarely changes, making it a static variation point.
  - The end-effector is a specific robot component utilized to handle items, such as a hand or vacuum pad. Some logistics robots for piece-picking can change their end-effector based on the items in the order list, making this variation point dynamic.
  - Sensors (such as cameras) are robot components used to detect items for picking and measure their size and position. The selection of sensors is done during equipment installation and rarely changes, making it a static variation point.
- Mission refers to the variability outside of logistics robots related to the tasks they perform.
  - Items are the products to be picked. As new items are released every day and their packaging (shape, color, weight) varies, robots must select the appropriate picking strategy, end-effector, and arm trajectory. This variation point is dynamic.
  - Item layout refers to the arrangement of items, for example, inside a shipping box. Since the item list is different for each order, the item layout changes for each order. The robot control software must design the layout and determine the order of item picking and placement. This variation point is dynamic.

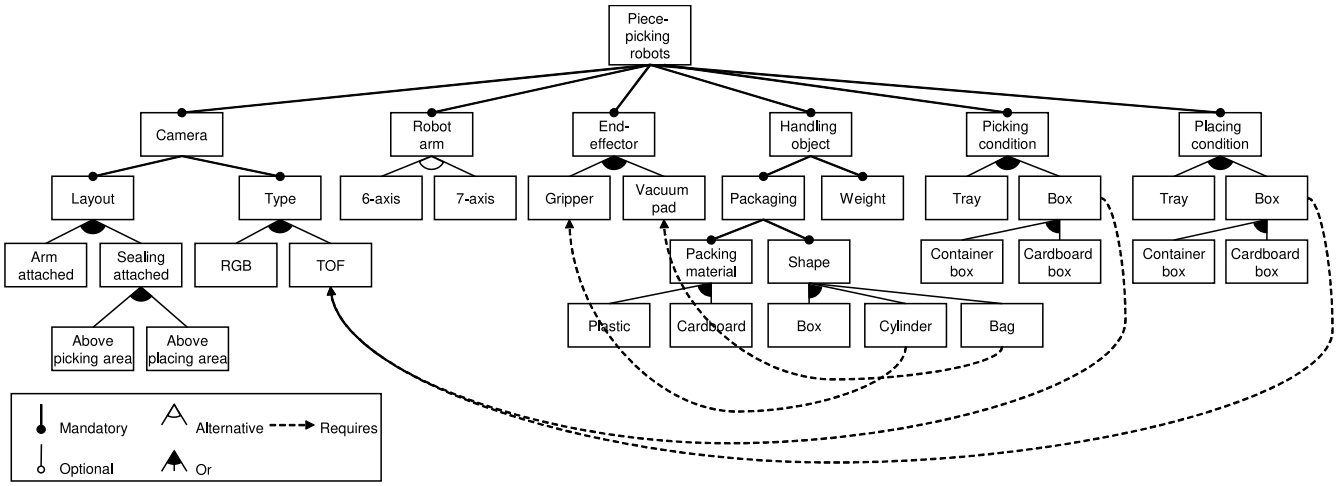

Fig. 2. Feature model for piece-picking robots.

We utilize feature-oriented variability modeling (Kang et al., 2002; Lee and Muthig, 2006; Apel et al., 2013) to clarify the variability of piece-picking robotic applications. Fig. 2 provides an overview of the feature model we use for piece-picking robots. As discussed in the Related Work section, there are two categories of variability in logistics robots: robot configuration and the product to be handled. It should be noted that the feature model is presented in an abstract manner due to confidentiality concerns. However, we aim to discuss the domain problem of logistics robots and share our experiences to the greatest extent possible.

In Fig. 2, the left side depicts the features of the robot configurations, including the camera, robotic arm, and end-effector. The camera is utilized to detect the object to be handled and recognize its shape, packaging, and points to be gripped or vacuumed. Depending on the application, the camera may be attached to the robotic arm and/or fixed above the picking area. In some applications, a camera also needs to be positioned above the placing area to detect empty space in a container for placement. Camera devices exhibit additional variability, e.g., detection type. Typically, a camera will be equipped with an RGB sensor (such as a CMOS sensor), and some applications additionally require a time-of-flight (ToF) sensor to measure the depth of the picking container and detect items in multi-tiered boxes. The robotic arm moves the position and angle of the end-effector. As we utilize multi-purpose robotic arms for our application, the variability primarily refers to the specific type of robotic arm (size, capacity, and number of axes).

The end-effector grips the product to be handled. Grippers enable robots to pick up and hold objects, while vacuum end-effectors are designed for handling fragile items, such as crisps in plastic bags. Depending on the application, the end-effector may be a gripper, a vacuum pad, or a hybrid of the two. The right side of Fig. 2 shows the variability of the product to be handled, the handling object, the picking condition, and the placing condition.

From the perspective of logistics robot control, packaging and weight are important variabilities and parameters when handling objects. For example, fragile objects (e.g., crisps in plastic bags) require a vacuum end-effector, while other packages that are weightier (e.g., green tea in PET bottles) require a gripper end-effector. As such, we have defined the package material and shape as major sub-features of the handling object package feature.

In addition to the product itself, the storage and shipping containers represent important variabilities for piece-picking robots. Depending on the logistics process and the customer, there is variability in how products are stored and shipped. For example, when a warehouse receives a product, it may arrive in a cardboard box, and the robot must depalletize it into a container box. Conversely, in the same warehouse, the robot may pick an item from a container and place it in another container box specified by the customer for shipping.

5.3. Defining variability mechanism

To achieve the desired variability in piece-picking robots, our study addressed two distinct categories of challenges related to variations. Firstly, we aimed to manage the variations in software for different components of the system with minimal effort. Secondly, we sought to address the necessity for modifications in abstract interfaces as the project progresses. Ideally, when system components exhibit variations (e.g., the picking condition and the end-effector mechanism type), we aim to provide code for each of these variations and independently select them to form a viable program. However, there remains code that is dependent on the combination of the choices made for the two components. Our objective is to minimize this type of code in an elegant manner. To achieve this, we utilized an approach that involves defining two types of abstractions (interface classes): one in which the concrete class is selected based on the type of a single system component, and another in which the class is selected based on the combination of the types of two or more components. The goal is to allow the former to handle most of the variations, while the latter addresses the hopefully small amount of remaining code.

In our study, we endeavored not to over-analyze the design problem upfront by defining numerous abstract interfaces to accommodate potential variations that may not be well understood at the time of design. Instead, we adopted an agile mindset, refraining from designing variations until they were needed and understood. Consequently, unlike conventional Product Line (PL) applications, the abstract interfaces defined in the system are subject to modification in future releases.

To achieve our concept, we utilize the strategy pattern (Gamma et al., 1994), which is a behavioral software design pattern (also known as the policy pattern) that enables the selection of an algorithm at runtime. Rather than implementing a single algorithm directly, code receives runtime instructions specifying which algorithm from a family of algorithms to use.

Given that logistics robots need to handle a variety of products, the robotic system must bind the variability for each product. As new products are added on a daily basis, we apply the strategy pattern as a mechanism for the late binding of the variability of pick-and-place tasks.

Fig. 3 shows the class diagram of the piece-picking robot system. In our design, logistics automation serves as the context for the strategy pattern. The pick-and-place task comprises the scanning strategy interface, the picking strategy interface, the placing strategy interface, and the hand motion strategy interface. Each strategy interface has a specific implementation to abstract the differences between variabilities. For example, the picking strategy interface includes the tray picking

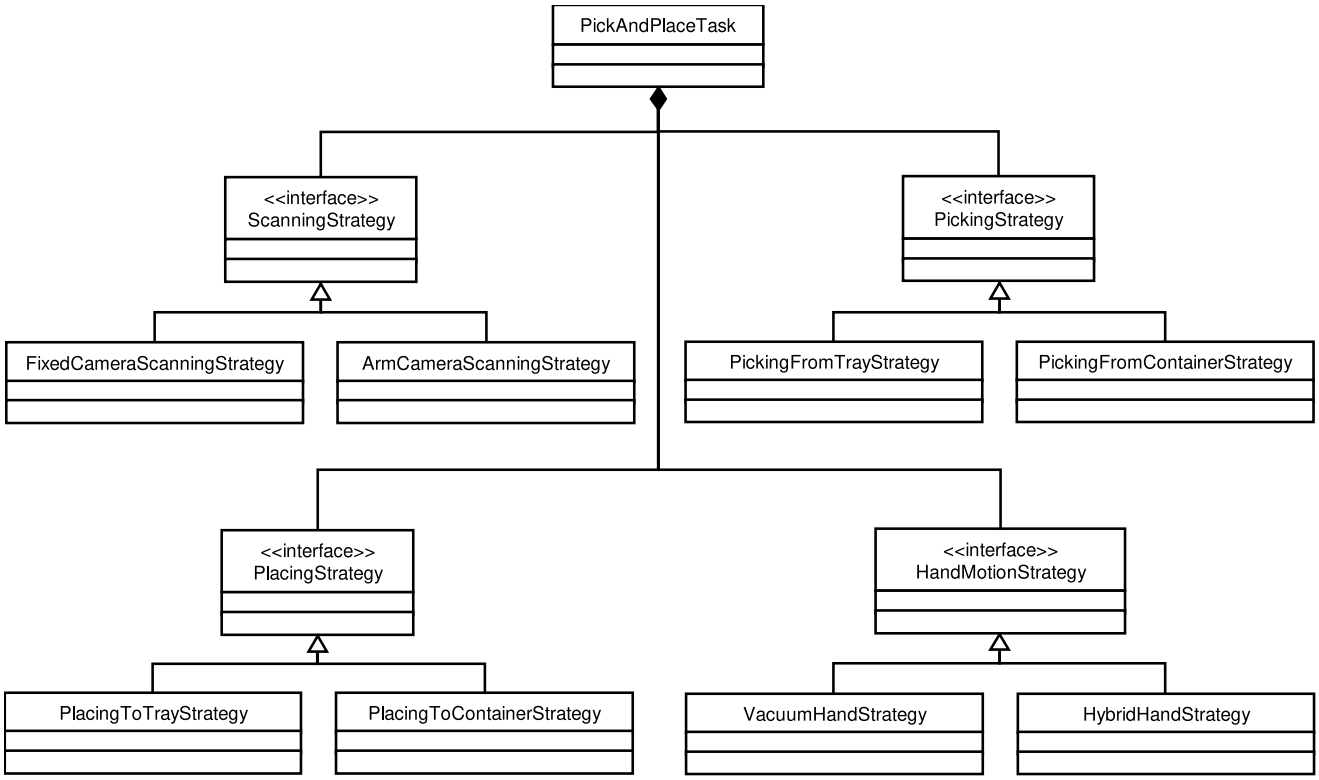

Fig. 3. Class diagram of pick-and-place task and strategies.

strategy and the container picking strategy. The context selects a strategy for execution based on information about the storage of the items. Similarly, the hand motion strategy interface includes the vacuum hand strategy and the hybrid hand strategy. Using the product package and weight as a basis, the robotic system selects the appropriate hand motion strategy.

To expand variability, new strategies may be added to the interface. For example, note that although the conveyor picking feature is extracted in Fig. 2, it is absent from Fig. 3. If a new customer requires conveyor picking, a new strategy can be added to expand the variability. Additionally, when new devices are developed, such as a soft grip hand, new strategies, such as a soft hand motion strategy, can be added.

5.4. Developing core & variable components

We developed the robot control system by utilizing the robot operating system (ROS) and implement the software components as nodes within the ROS environment. A node is an executable that utilizes ROS to communicate with other nodes. Briefly, the ROS infrastructure facilitates the efficient implementation of the variation mechanism, thus enabling developers to reuse software components as ROS nodes. The behavior of the implementation is explained in the following subsection.

5.5. Deriving different applications

Finally, pick-and-place tasks are derived for each product to be handled. Fig. 4 depicts a use case in which a robot picks two objects - a bag of crisps in a foil bag stored on a tray and a box of cookies in a cardboard box - and places them into a container box for shipping.

The pick-and-place manager activates two tasks, one for the foil bag and one for the cardboard box, and each task selects its strategy on the basis of the context. As shown in Fig. 4, the task for the foil bag selects the appropriate strategies, such as the tray picking strategy and the vacuum pad hand motion strategy, and the task for the cookie box binds the strategy best suited for the cardboard box.

To improve the throughput, pick-and-place tasks are executed in parallel as multi-tasks. We define relationships between actions and robot devices; for example, the fixed camera scanning strategy uses the fixed camera. With this information as a basis, as soon as the predecessor task frees up resources, the successor task can initiate its action, such as scanning.

6. Evaluation

To evaluate the effectiveness of our variability management approach for logistics robotic systems, we conducted a series of interviews with key stakeholders involved in the development and deployment of the target robot application. Specifically, we interviewed the lead architect of the robot application and an expert programmer who has been deeply involved in the implementation and maintenance of the robot control software. These interviews provided valuable insights into the practical benefits and challenges of our approach.

6.1. Interview with the lead architect

The lead architect, with over 20 years of experience in the software industry, has a broad background spanning from enterprise IT architecture to robotic control system architecture. As an expert in software engineering, his extensive experience is highly relevant to the challenges addressed by our variability management approach. During the interview, he highlighted several key benefits of the approach. One of the most significant advantages was the ability to dynamically generate variations in robot behavior based on the binding results of the environment and mission (picking items) variability models. This capability allowed the system to adapt to different warehouse layouts, lighting conditions, and other environmental factors without requiring extensive manual reconfiguration. The architect noted that this flexibility was particularly valuable in logistics environments where conditions

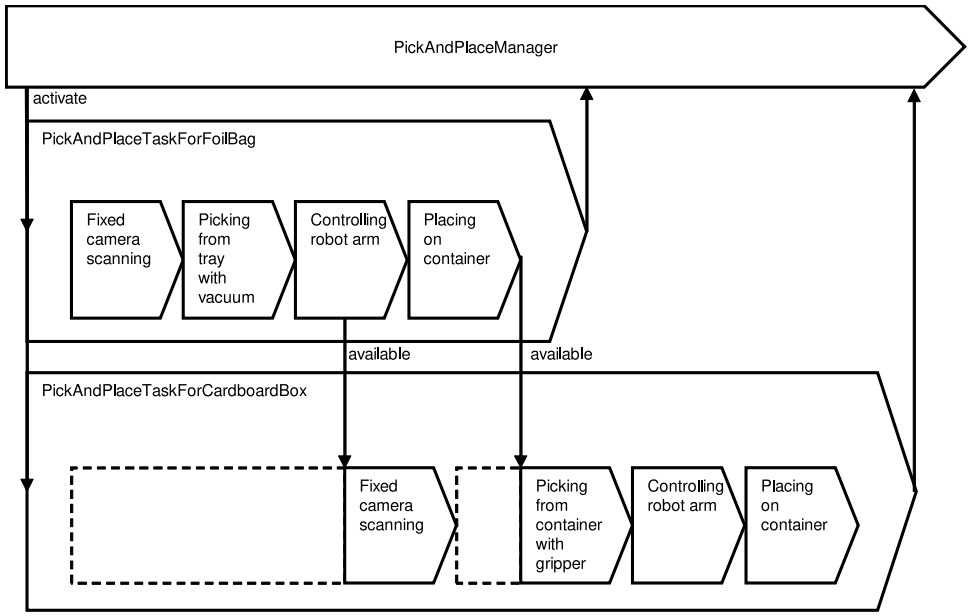

Fig. 4. Overview of behavior of pick-and-place tasks.

can change frequently and unpredictably.

Moreover, the architect emphasized that the approach significantly reduced the effort required for configuring the robot application. By leveraging the variability models, the system could automatically select the appropriate hardware and software components for each specific scenario. This automation not only saved time but also minimized the risk of human error during the configuration process. The architect estimated that the configuration effort was reduced significantly, allowing the team to focus more on developing new features and improving system performance.

6.2. Interview with the expert programmer

The expert programmer, with around 10 years of industry experience, specializes in robotics and industrial control systems. His expertise is crucial in understanding the technical aspects of the variability management approach, making his experience highly pertinent to the evaluation of our methodology. He confirmed that the use of feature-oriented variability modeling and the strategy pattern were highly effective in managing the complexity of the robot control software. The programmer explained that the strategy pattern allowed for the late binding of variability, enabling the system to dynamically select the most appropriate strategies for scanning, picking, placing, and hand motion based on the context of the items to be handled.

The programmer also noted that the approach facilitated the integration of new hardware components, such as novel end-effectors and sensors, into the system. By defining new strategies and adding them to the existing interfaces, the team could easily extend the system's capabilities without disrupting the core functionality. This modularity and extensibility were crucial for keeping the system up-to-date with the latest technological advancements in robotics.

6.3. Identified benefits

Based on the interviews, we identified the following key benefits of our variability management approach:

- **Dynamic Generation of Robot Behavior Variations**: The ability to dynamically generate variations in robot behavior based on the binding results of the environment and mission variability models was highly effective. This capability allowed the system to adapt to different conditions and requirements with minimal manual intervention.
- **Reduced Configuration Effort**: The approach significantly reduced the effort required for configuring the robot application. By automating the selection of hardware and software components, the system minimized the need for manual reconfiguration and reduced the risk of human error.
- **Improved Flexibility and Adaptability**: The use of feature-oriented variability modeling and the strategy pattern provided a flexible and adaptable framework for managing the complexity of the robot control software. This flexibility was particularly valuable in dynamic and unpredictable logistics environments.
- **Enhanced Modularity and Extensibility**: The approach facilitated the integration of new hardware components and the extension of system capabilities. By defining new strategies and adding them to the existing interfaces, the team could easily incorporate new technologies and improve system performance.

6.4. Identified challenges

While the evaluation highlighted several benefits, it also identified areas for future work and challenges that need to be addressed:

- **Generalization of Variability Modeling for Autonomous Systems**: The classical SPL approach primarily targets hardware variations of the target system. However, it is not clear how to model the variability of recent autonomous systems, such as environmental and mission variability. To apply this approach to other autonomous robotics, a more generalized approach is needed. This involves developing methodologies that can effectively capture and manage the dynamic and complex variability inherent in autonomous systems.
- **Integration with Robot Software Platforms**: While software platforms for robotic applications, such as the Robot Operating System (Macenski et al., 2022), provide the concept of configuration parameters, they lack mechanisms to manage the relationships between configuration parameters required for variability management. As autonomous robotics transition from the research stage to practical applications, there is a need for the integration of robot software and variability management. This

integration would enable more efficient and effective management of the complex configurations and dependencies in robotic systems.

In conclusion, the evaluation of our variability management approach demonstrated its effectiveness in reducing configuration effort, improving flexibility and adaptability, and enhancing modularity and extensibility. The insights gained from the interviews with key stakeholders will guide future work in further refining and extending the approach to meet the evolving needs of logistics robotic systems. By addressing the identified challenges and pursuing the outlined areas for future work, we aim to continue to improve the effectiveness and applicability of variability management in autonomous robotic systems.

7. Discussion and future work

In this section, the benefits and future work are discussed.

To effectively manage variability in logistic robotic systems, a variability management approach is beneficial. This is due to the complex nature of the logistics business, which involves variability in products, operational environments, and the robots themselves. Modeling these variabilities aids in understanding them across various stakeholders, including software developers, facility engineers, robot engineers, requirement engineers, and logistic operators (customers).

Utilizing software design (such as the factory pattern) and implementation can improve system development cost and time. The variability model allows for the design of new systems by selecting features and identifying new variation points for specific robotic systems. Furthermore, the factory pattern enables the addition of new functionalities, such as concrete implementation for innovative end-effector devices, in an efficient manner.

On the other hand, the management of variability in real products is a complex and iterative process. Although the abstracted variability management approach is depicted as a straightforward process in Section 5, several challenges arise when managing variability. For example, the variability of logistics operations is often uncontrollable. This is particularly true for food and cosmetic packaging where differences in packaging are key features of new products. Furthermore, with the accelerating development of robotics technology, novel devices such as hands and sensors are required. As such, an incremental approach to variability management is necessary. Once the robot control software has been implemented, variability analysis and component development must be conducted continuously.

We have also identified several areas for future work, including the need for generalization of variability modeling for autonomous systems and for integration with robot software platforms. Additionally, the integration of generative AI and software product lines (SPL) has the potential to enhance the ability of service robots to understand their context and configure themselves accordingly. As it is not feasible to design for all possible contexts or to define the relationships between context and variability in advance, AI-augmented robot operation approaches have been proposed (Pyo et al., 2022; Garello et al., 2021; Vemprala et al., 2023). The integration of generative AI to automatically transform context into a variability model will be a key technology in the robotics domain.

8. Conclusion

In this paper, we presented an experience report on the application of variability management to logistics robot systems. Our approach includes the use of feature-oriented variability analysis. Throughout our report, we discuss the characteristics and drivers of variability in robotic applications, the practice of variability management in logistics robots, and the dynamic binding mechanism of variability based on the context of items to be handled. Our results contribute to the empirical understanding of the specific characteristics and challenges of the logistics robotics domain with respect to variability. We anticipate that our findings will support practitioners and researchers in raising awareness of the effectiveness of variability management and guide future research in this area.

CRediT authorship contribution statement

Kentaro Yoshimura: Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Yuta Yamauchi:** Writing – review & editing, Software, Methodology. **Hideo Takahashi:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Kentaro Yoshimura has patent pending to Hitachi, Ltd. Corresponding author's research group has a joint research project with Fraunhofer IESE, Germany. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Kentaro Yoshimura is a chief researcher at Hitachi, Ltd. His research interests include software product line engineering for industrial control systems and applications. Yoshimura has a PhD in information science and technology from Osaka University.

Yuta Yamauchi is a senior engineer at Hitachi, Ltd. He is an expert of software platform for robotic applications.

Hideo Takahashi is a chief engineer at Hitachi, Ltd. He is an expert of software architectures including robotic applications.